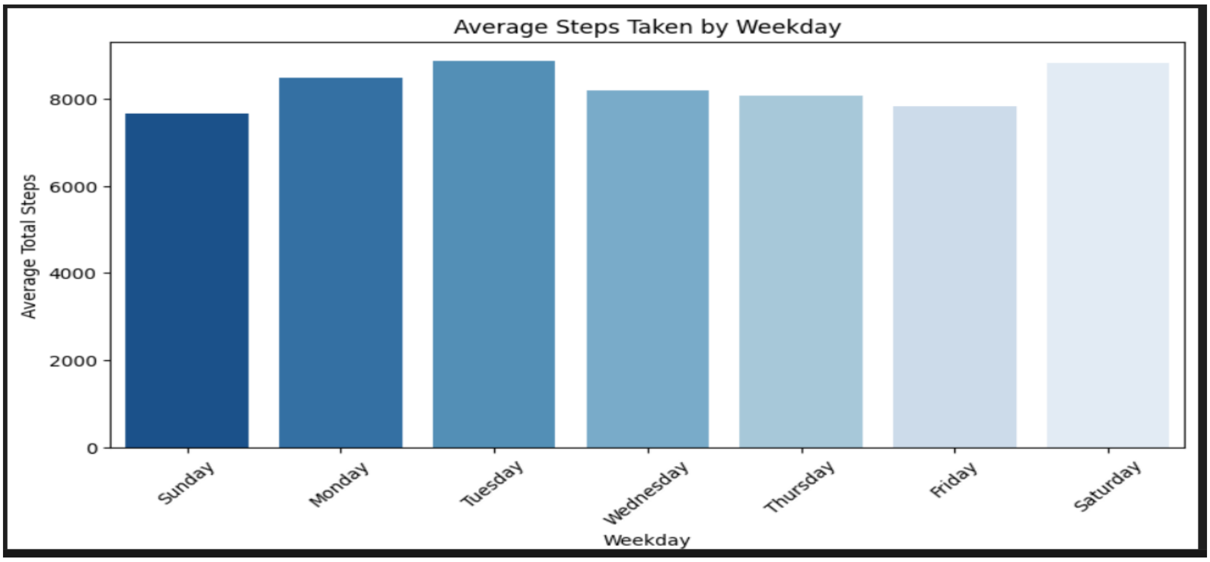
STRAVA FITNESS SCREENSHOTS

Python Visualization -

Strava EDA Analysis

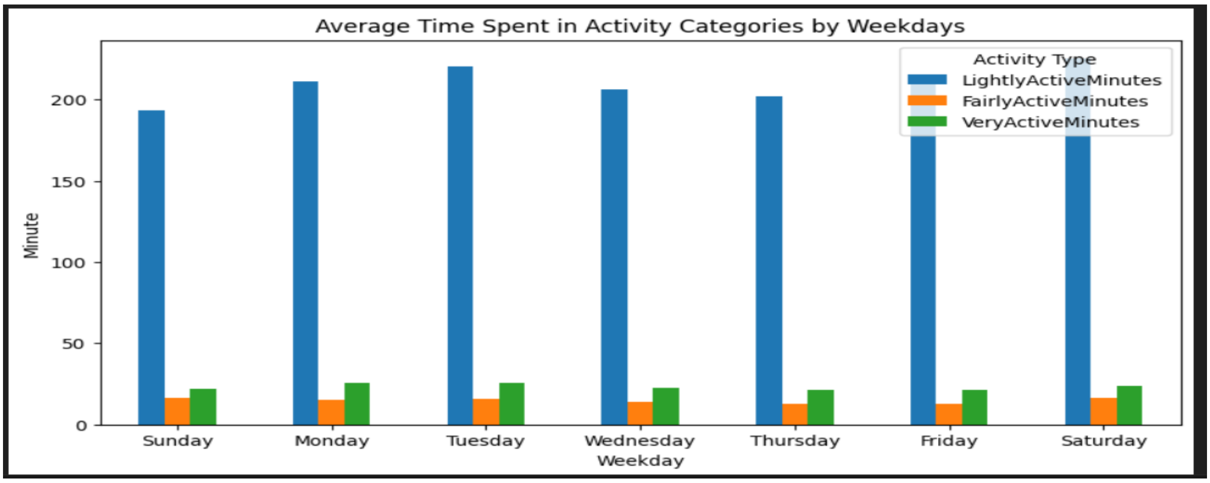
```python
# --- Daily Activity ---

# Convert ActivityDate to datetime and extract weekday
activity_df['ActivityDate'] = pd.to_datetime(activity_df['ActivityDate'])
activity_df['Weekday'] = activity_df['ActivityDate'].dt.day_name()


# Group by weekday
steps_by_day = activity_df.groupby('Weekday')['TotalSteps'].mean().reindex([
    'Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday'
])

# Plot
plt.figure(figsize=(10,5))
sns.barplot(x=steps_by_day.index, y=steps_by_day.values, palette='Blues_r')
plt.title('Average Steps Taken by Weekday')
plt.ylabel('Average Total Steps')
plt.xticks(rotation=45)
plt.show()
```

✓ 0.4s



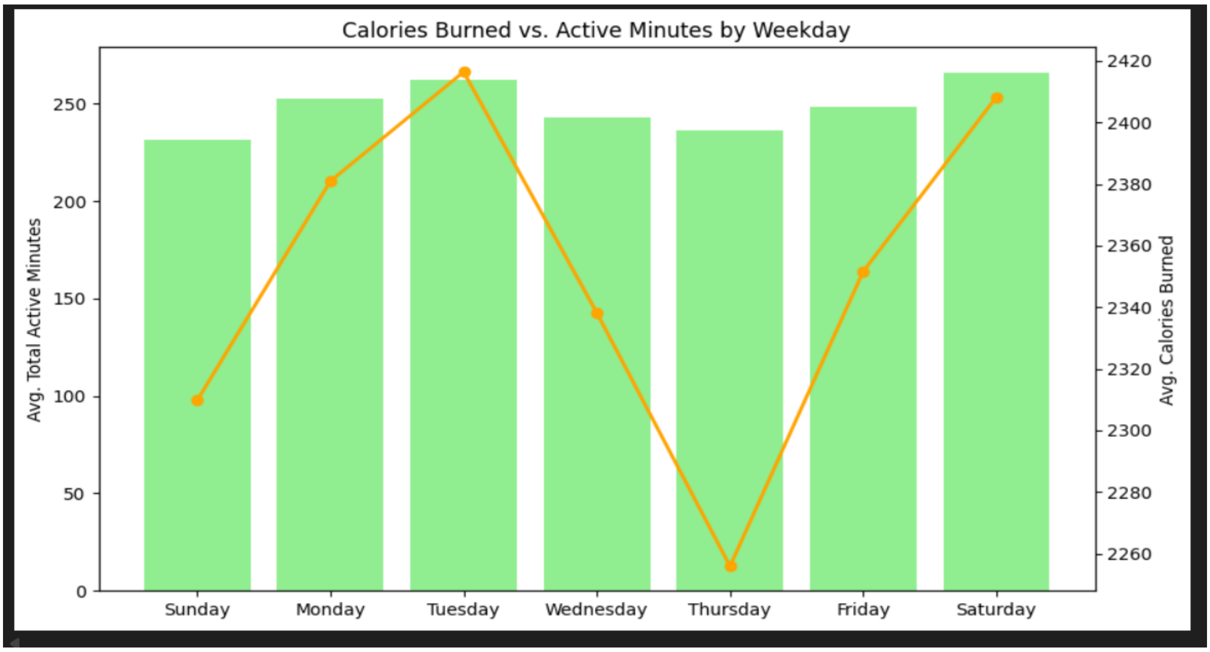

```python
activity_minute = activity_df.groupby('Weekday')[[
    'LightlyActiveMinutes', 'FairlyActiveMinutes', 'VeryActiveMinutes'
]].mean().reindex(['Sunday','Monday','Tuesday', 'Wednesday','Thursday','Friday','Saturday'])

#Plot
activity_minute.plot(kind = 'bar', figsize=(10,5))
plt.title('Average Time Spent in Activity Categories by Weekdays')
plt.ylabel('Minute')
plt.xticks(rotation=0)
plt.legend(title='Activity Type')
plt.show()
```

✓ 0.2s



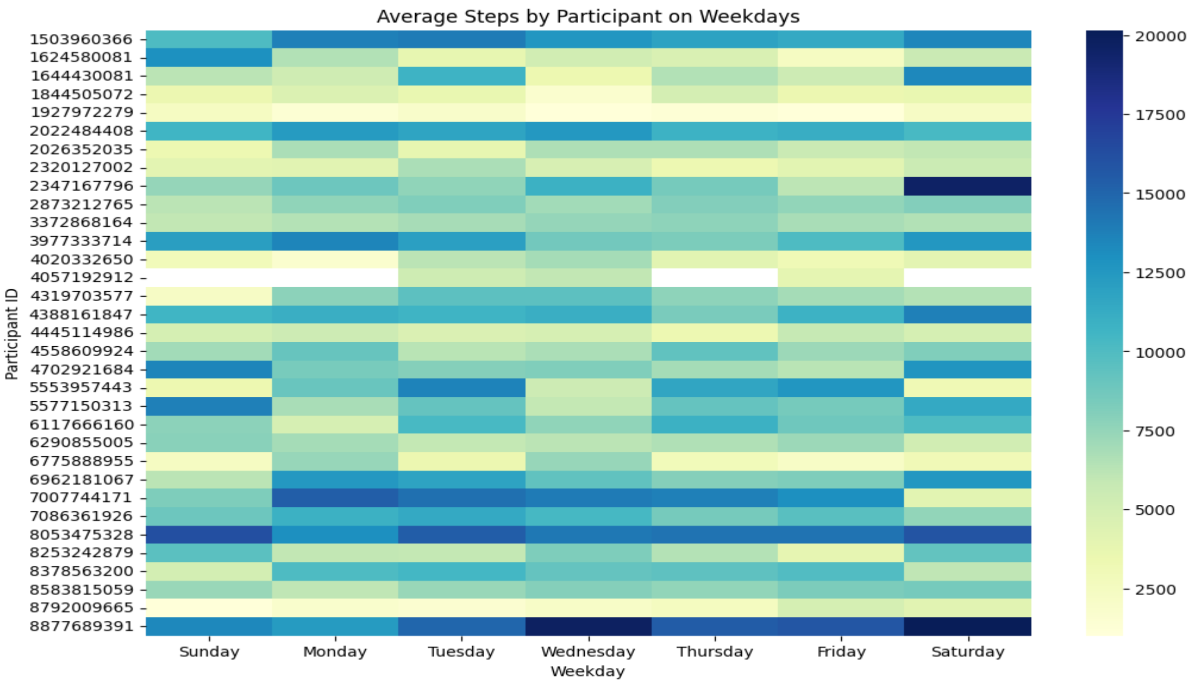

```python
# Create a new column for total active minutes
activity_df['TotalActiveMinutes'] = activity_df['LightlyActiveMinutes'] + activity_df['FairlyActiveMinutes'] + activity_df['VeryActiveMinutes']

# Group and average by weekday
calories_df =  activity_df.groupby('Weekday')[['TotalActiveMinutes', 'Calories']].mean().reindex([
    'Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday'])

# Plot dual-axis chart
fig, ax1 = plt.subplots(figsize=(10,6))
ax1.bar(calories_df.index, calories_df['TotalActiveMinutes'], color='lightgreen')
ax1.set_ylabel('Avg. Total Active Minutes')

ax2 = ax1.twinx()
ax2.plot(calories_df.index, calories_df['Calories'], color='orange', marker='o', linewidth=2)
ax2.set_ylabel('Avg. Calories Burned')

plt.title('Calories Burned vs. Active Minutes by Weekday')
plt.xticks(rotation=45)
plt.show()
```

✓ 0.2s



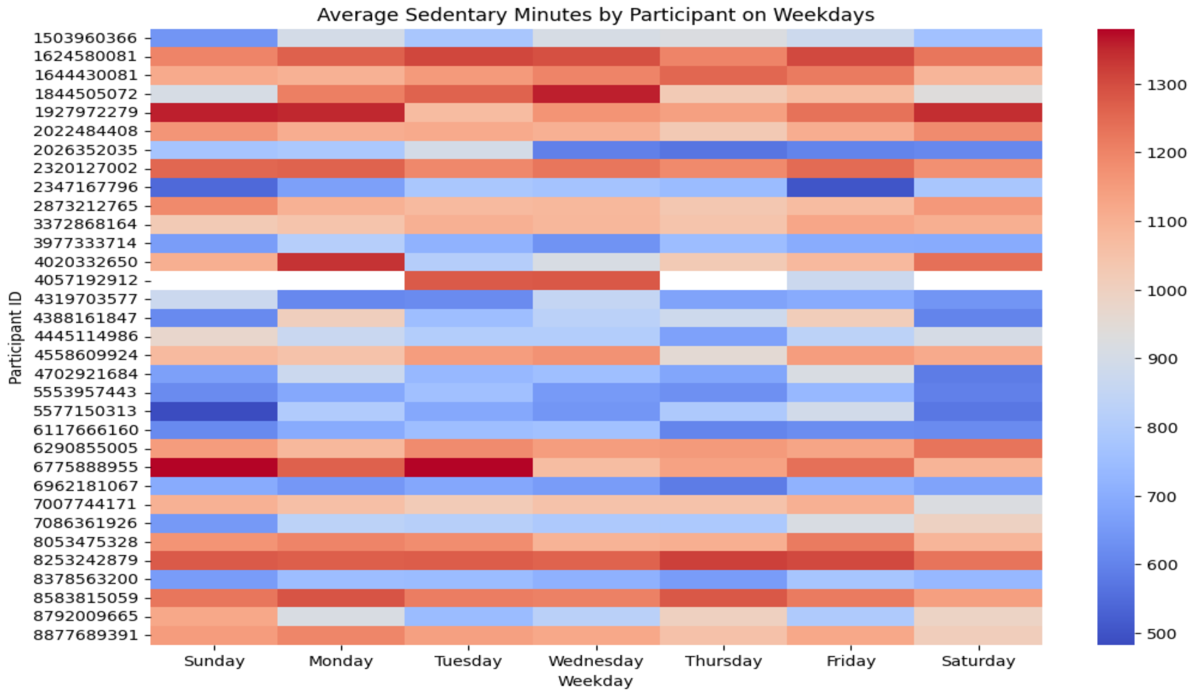
```python
#activity_df['ActivityDate'] = pd.to_datetime(activity_df['ActivityDate'])
#activity_df['Weekday'] = activity_df['ActivityDate'].dt.day_name()

# Pivot for heatmap
steps_heatmap = activity_df.pivot_table(
    index='Id', columns='Weekday', values='TotalSteps', aggfunc='mean'
).reindex(columns=['Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday'])

# Plot heatmap
plt.figure(figsize=(12, 8))
sns.heatmap(steps_heatmap, cmap='YlGnBu')
plt.title('Average Steps by Participant on Weekdays')
plt.xlabel('Weekday')
plt.ylabel('Participant ID')
plt.show()
```

✓ 0.4s

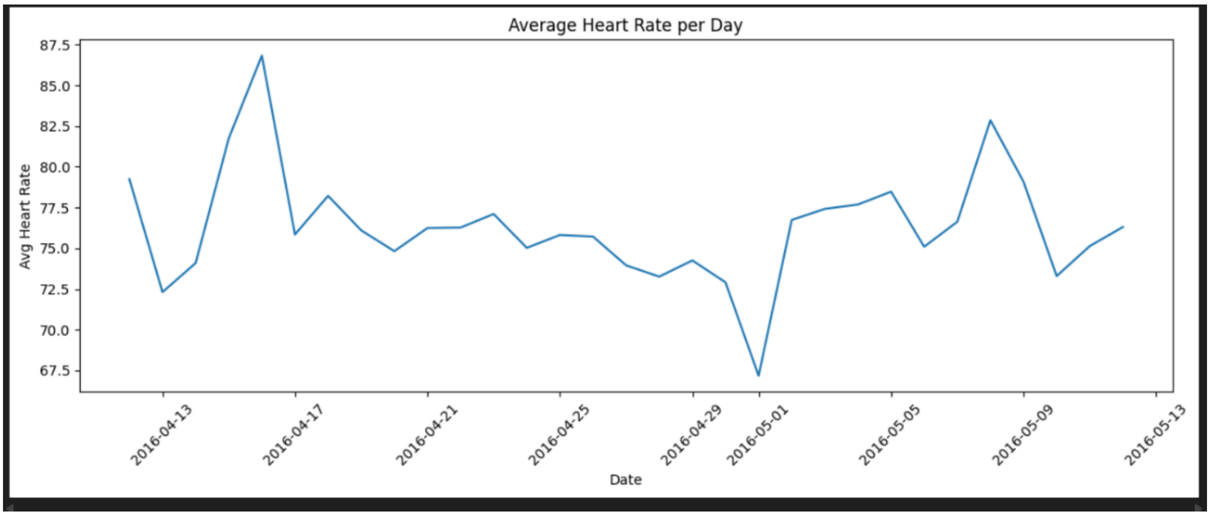



```python
# Pivot for sedentary heatmap
sedentary_heatmap = activity_df.pivot_table(
    index='Id', columns='Weekday', values='SedentaryMinutes', aggfunc='mean'
).reindex(columns=['Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday'])

# Plot heatmap
plt.figure(figsize=(12, 8))
sns.heatmap(sedentary_heatmap, cmap='coolwarm')
plt.title('Average Sedentary Minutes by Participant on Weekdays')
plt.xlabel('Weekday')
plt.ylabel('Participant ID')
plt.show()
```

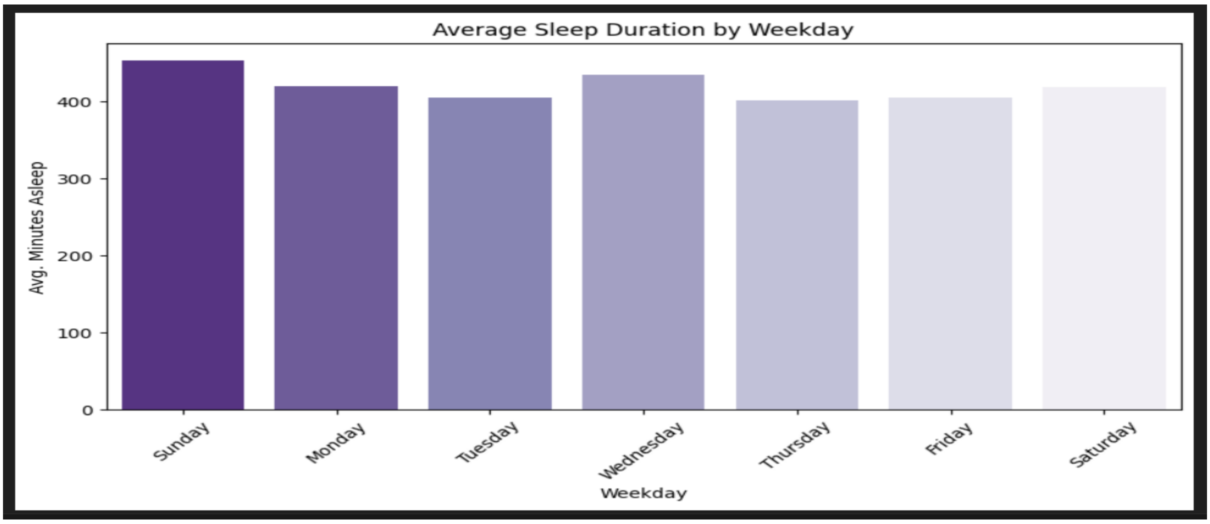
✓ 0.4s



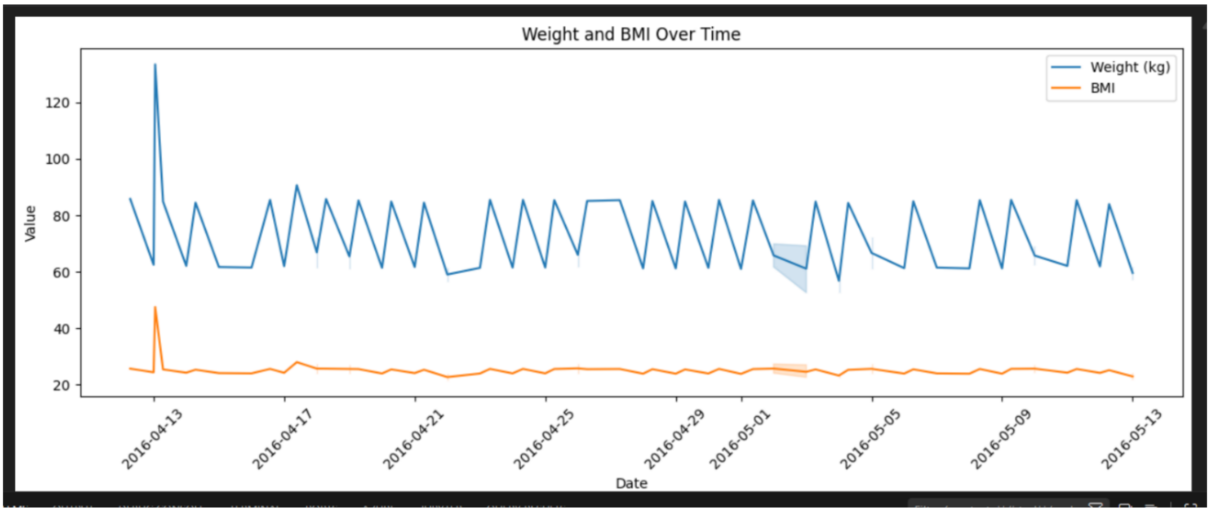

```python
# 1. Average Heart Rate per Day
avg_daily_hr = heart_df.groupby('Date')['Value'].mean()

plt.figure(figsize=(12, 5))
avg_daily_hr.plot()
plt.title('Average Heart Rate per Day')
plt.xlabel('Date')
plt.ylabel('Avg Heart Rate')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```python
# --- Sleep Day ---

# Convert date column to datetime and extract weekday
sleepDay_df['SleepDay'] = pd.to_datetime(sleepDay_df['SleepDay'])
sleepDay_df['Weekday'] = sleepDay_df['SleepDay'].dt.day_name()



# Group by weekday and calculate average sleep
avg_sleep = sleepDay_df.groupby('Weekday')['TotalMinutesAsleep'].mean().reindex([
    'Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday'
])

# Plot
plt.figure(figsize=(10,5))
sns.barplot(x=avg_sleep.index, y=avg_sleep.values, palette='Purples_r')
plt.title('Average Sleep Duration by Weekday')
plt.ylabel('Avg. Minutes Asleep')
plt.xticks(rotation=45)
plt.show()
```



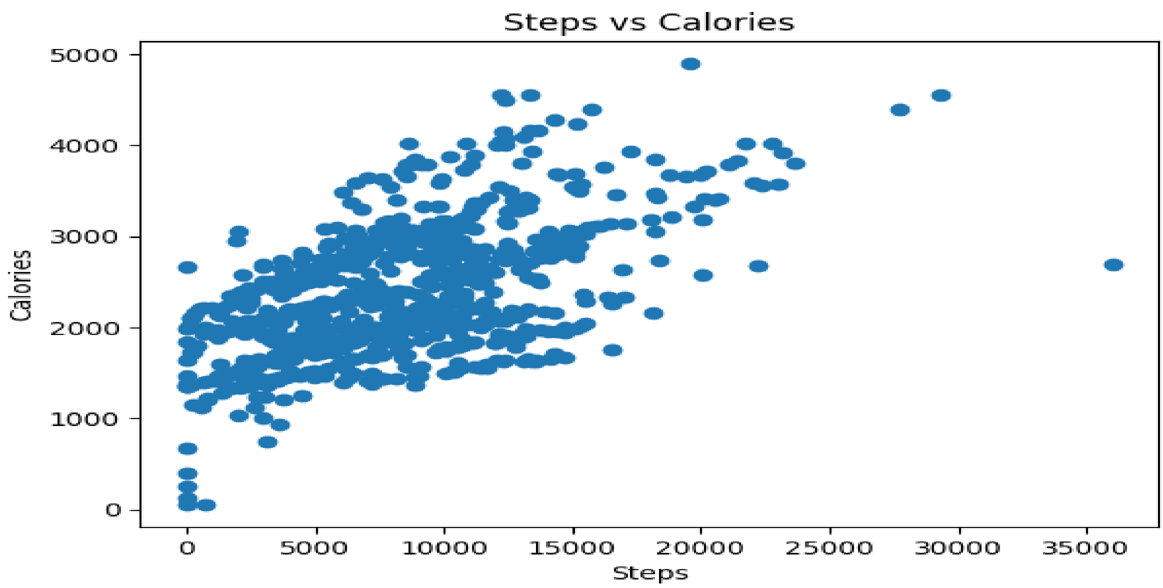

```python
# --- Weight Log Info ---

# Convert the 'Date' column to datetime
weightInfo_df['Date'] = pd.to_datetime(weightInfo_df['Date'])

# Categorize BMI into health categories
def categorize_bmi(bmi):
    if bmi < 18.5:
        return 'Underweight'
    elif 18.5 <= bmi < 25:
        return 'Normal'
    elif 25 <= bmi < 30:
        return 'Overweight'
    else:
        return 'Obese'

weightInfo_df['BMI_Category'] = weightInfo_df['BMI'].apply(categorize_bmi)

# 1. Weight and BMI Over Time
plt.figure(figsize=(12, 5))
sns.lineplot(data=weightInfo_df, x='Date', y='WeightKg', label='Weight (kg)')
sns.lineplot(data=weightInfo_df, x='Date', y='BMI', label='BMI')
plt.title('Weight and BMI Over Time')
plt.xlabel('Date')
plt.ylabel('Value')
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Merged EDA Analysis

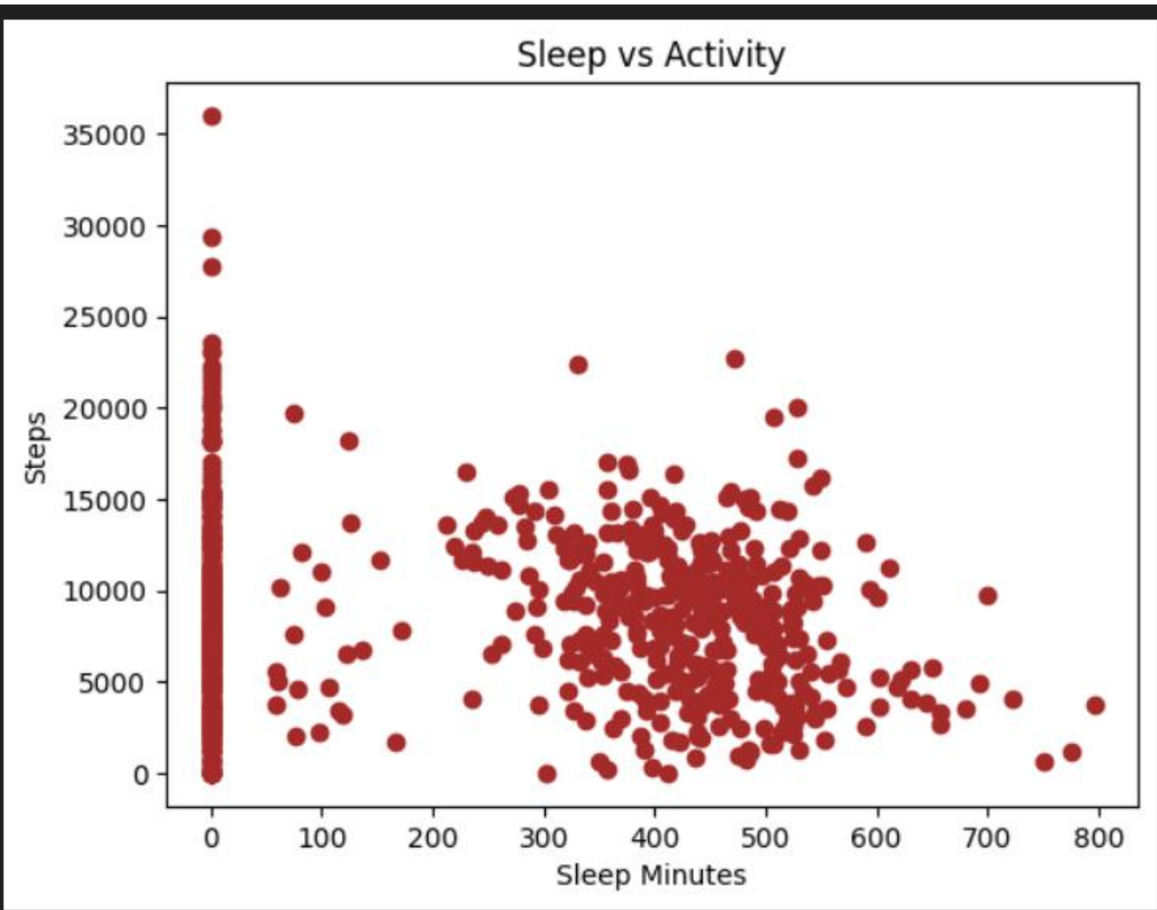
```python
# Steps vs Calories

plt.scatter(merged['TotalSteps'], merged['Calories'])
plt.xlabel("Steps")
plt.ylabel("Calories")
plt.title("Steps vs Calories")
plt.show()
```

23] ✓ 0.1s

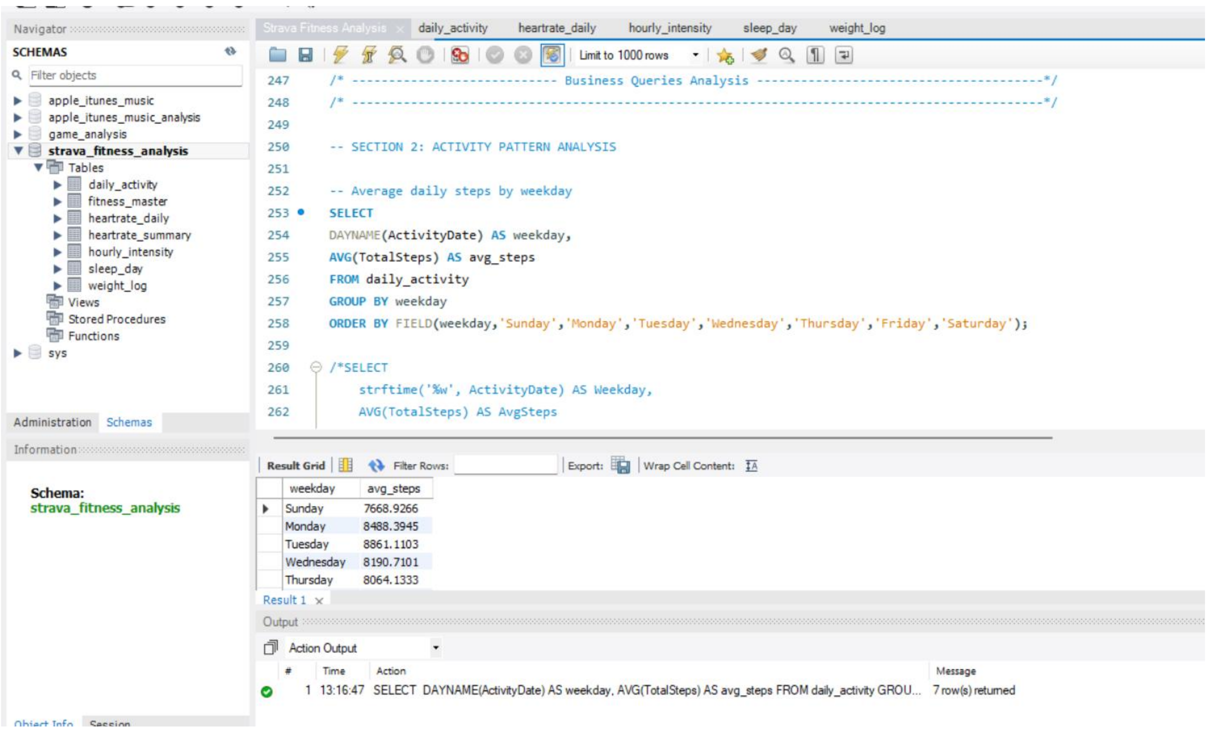


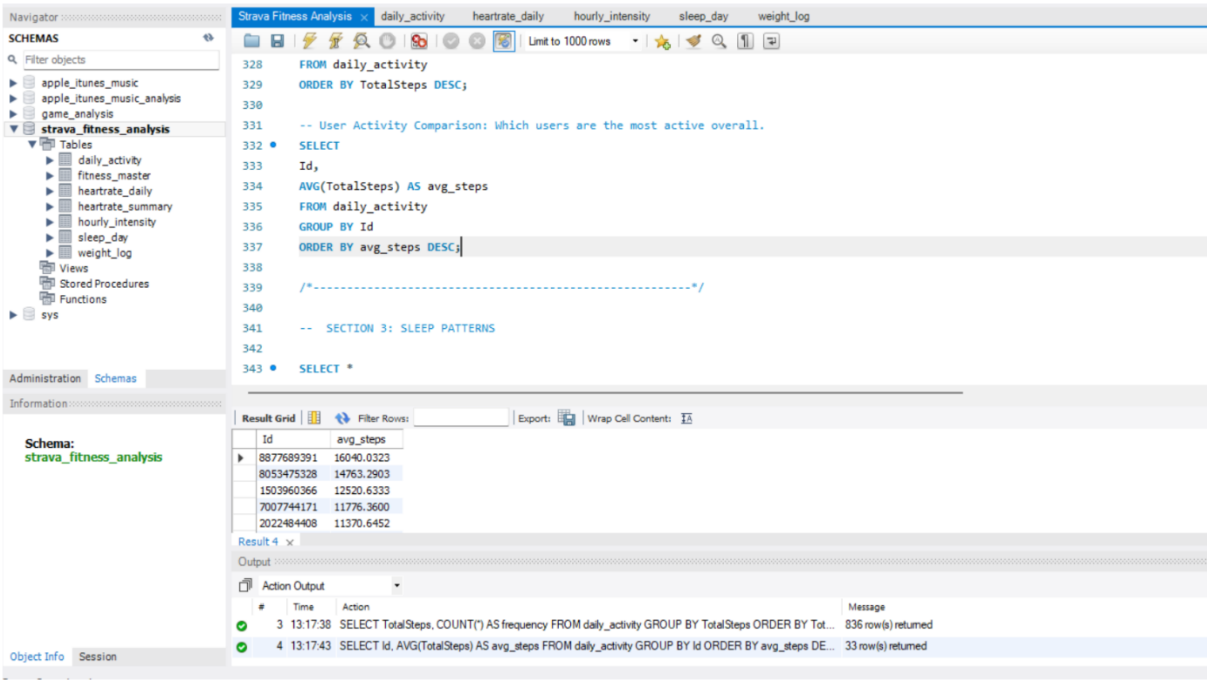
```
#Sleep vs Steps

plt.scatter(merged['TotalMinutesAsleep'], merged['TotalSteps'], color="brown")
plt.xlabel("Sleep Minutes")
plt.ylabel("Steps")
plt.title("Sleep vs Activity")
plt.show()
```

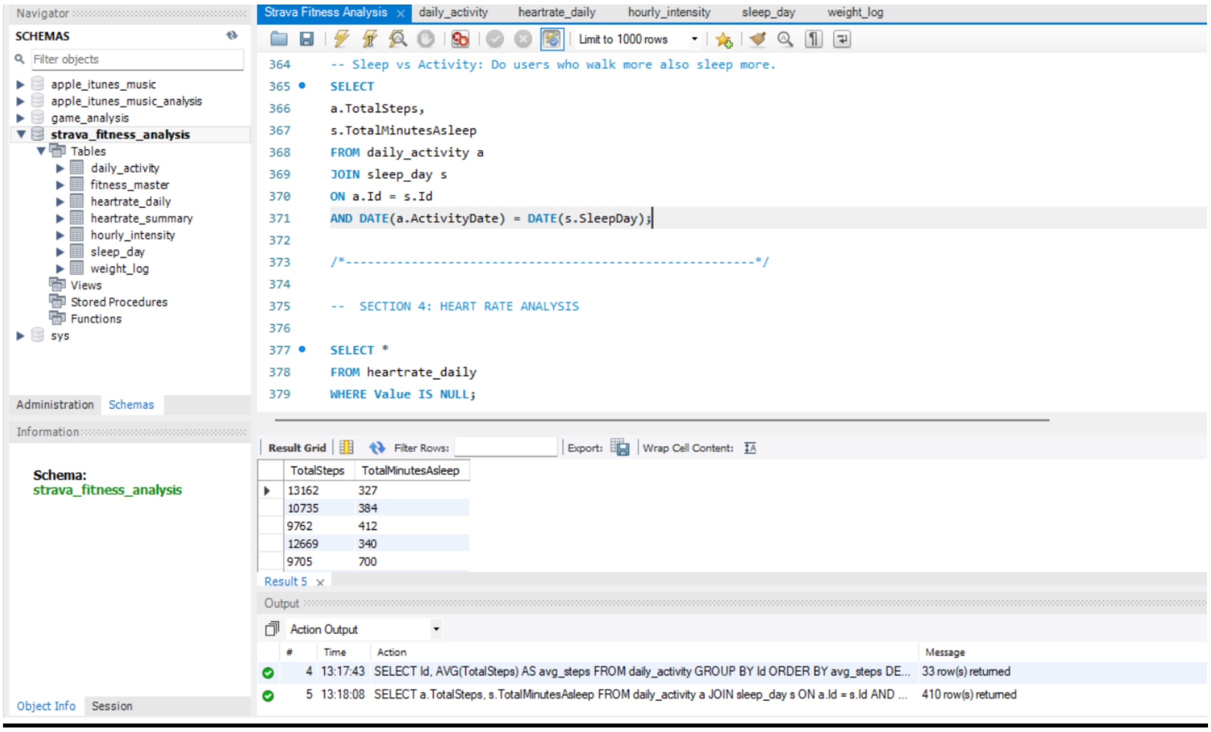
✓ 0.1s

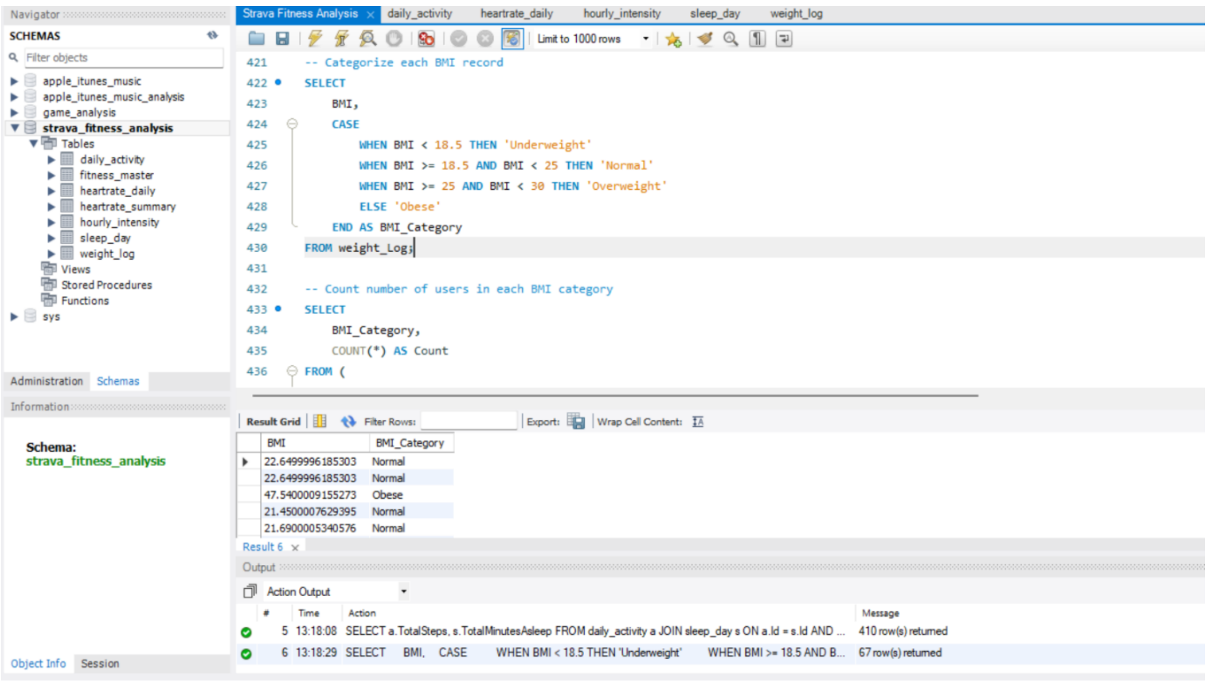


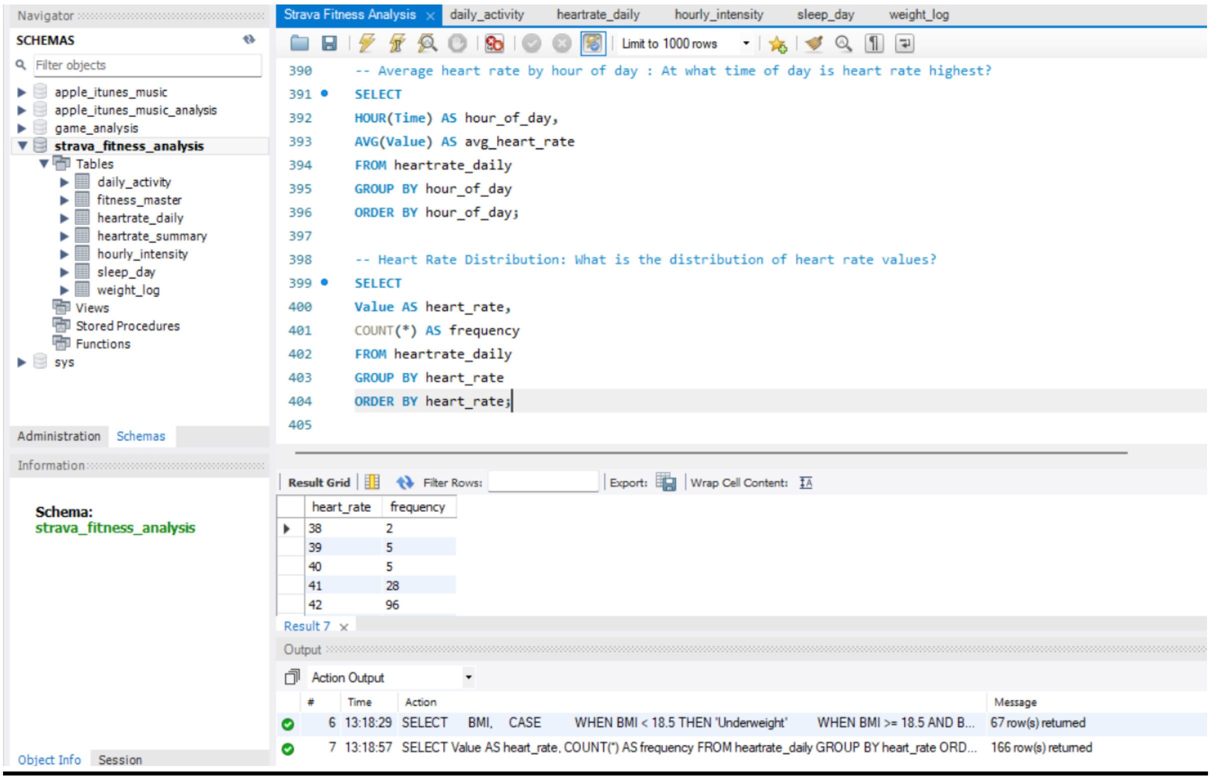
MySQL –


Navigator
SCHEMAS
apple_itunes_music
apple_itunes_music_analysis
game_analysis
strava_fitness_analysis
Tables
daily_activity
fitness_master
heartrate_daily
heartrate_summary
hourly_intensity
sleep_day
weight_log
Views
Stored Procedures
Functions
sys
Strava Fitness Analysis
Limit to 1000 rows
247 /* --------------------------- Business Queries Analysis -------------------------------------*/
248 /* ---*/
250 -- SECTION 2: ACTIVITY PATTERN ANALYSIS
252 -- Average daily steps by weekday
253 SELECT
254 DAYNAME(ActivityDate) AS weekday,
255 AVG(TotalSteps) AS avg_steps
256 FROM daily_activity
257 GROUP BY weekday
258 ORDER BY FIELD(weekday,'Sunday','Monday','Tuesday','Wednesday','Thursday','Friday','Saturday');
260 /*SELECT
261 strftime('%w', ActivityDate) AS Weekday,
262 AVG(TotalSteps) AS AvgSteps
Administration Schemas
Information
Schema:
strava_fitness_analysis
Result Grid Filter Rows: Export: Wrap Cell Content:
weekday avg_steps
Sunday 7668.9266
Monday 8488.3945
Tuesday 8861.1103
Wednesday 8190.7101
Thursday 8064.1333
Result 1
Output
Action Output
Time Action Message
1 13:16:47 SELECT DAYNAME(ActivityDate) AS weekday, AVG(TotalSteps) AS avg_steps FROM daily_activity GROU... 7 row(s) returned


Navigator
SCHEMAS
apple_itunes_music
apple_itunes_music_analysis
game_analysis
strava_fitness_analysis
Tables
daily_activity
fitness_master
heartrate_daily
heartrate_summary
hourly_intensity
sleep_day
weight_log
Views
Stored Procedures
Functions
sys
Strava Fitness Analysis
Limit to 1000 rows
328 FROM daily_activity
329 ORDER BY TotalSteps DESC;
331 -- User Activity Comparison: Which users are the most active overall.
332 SELECT
333 Id,
334 AVG(TotalSteps) AS avg_steps
335 FROM daily_activity
336 GROUP BY Id
337 ORDER BY avg_steps DESC;
339 /*---*/
341 -- SECTION 3: SLEEP PATTERNS
343 SELECT *
Administration Schemas
Information
Schema:
strava_fitness_analysis
Result Grid Filter Rows: Export: Wrap Cell Content:
Id avg_steps
8877689391 16040.0323
8053475328 14763.2903
1503960366 12520.6333
7007744171 11776.3600
2022484408 11370.6452
Result 4
Output
Action Output
Time Action Message
3 13:17:38 SELECT TotalSteps, COUNT(*) AS frequency FROM daily_activity GROUP BY TotalSteps ORDER BY Tot... 836 row(s) returned
4 13:17:43 SELECT Id, AVG(TotalSteps) AS avg_steps FROM daily_activity GROUP BY Id ORDER BY avg_steps DE... 33 row(s) returned
Object Info Session

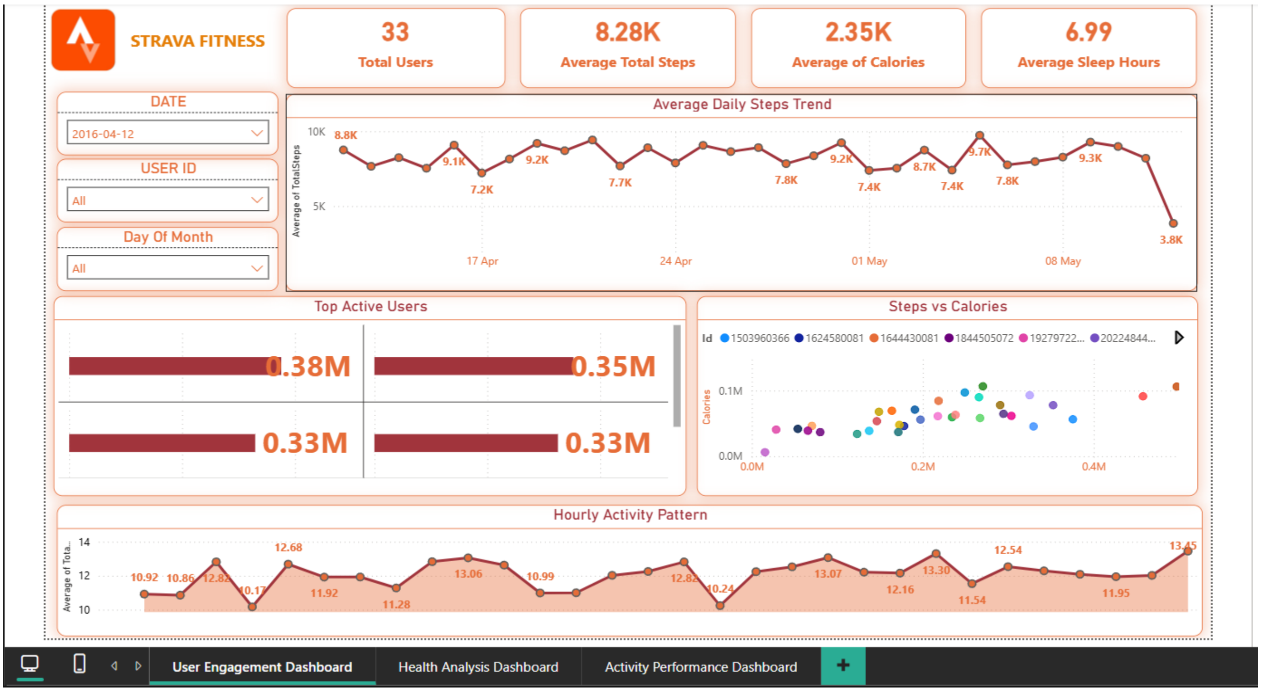

Navigator
SCHEMAS
Filter objects
apple_itunes_music
apple_itunes_music_analysis
game_analysis
strava_fitness_analysis
Tables
daily_activity
fitness_master
heartrate_daily
heartrate_summary
hourly_intensity
sleep_day
weight_log
Views
Stored Procedures
Functions
sys
Administration
Schemas
Information
Schema:
strava_fitness_analysis
Object Info
Session
Strava Fitness Analysis
daily_activity
heartrate_daily
hourly_intensity
sleep_day
weight_log
Limit to 1000 rows
-- Sleep vs Activity: Do users who walk more also sleep more.
SELECT
a.TotalSteps,
s.TotalMinutesAsleep
FROM daily_activity a
JOIN sleep_day s
ON a.Id = s.Id
AND DATE(a.ActivityDate) = DATE(s.SleepDay);
/*---*/
-- SECTION 4: HEART RATE ANALYSIS
SELECT *
FROM heartrate_daily
WHERE Value IS NULL;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
TotalSteps TotalMinutesAsleep
13162 327
10735 384
9762 412
12669 340
9705 700
Result 5
Output
Action Output
Time Action Message
4 13:17:43 SELECT Id, AVG(TotalSteps) AS avg_steps FROM daily_activity GROUP BY Id ORDER BY avg_steps DE... 33 row(s) returned
5 13:18:08 SELECT a.TotalSteps, s.TotalMinutesAsleep FROM daily_activity a JOIN sleep_day s ON a.Id = s.Id AND ... 410 row(s) returned


Navigator
SCHEMAS
Filter objects
apple_itunes_music
apple_itunes_music_analysis
game_analysis
strava_fitness_analysis
Tables
daily_activity
fitness_master
heartrate_daily
heartrate_summary
hourly_intensity
sleep_day
weight_log
Views
Stored Procedures
Functions
sys
Administration
Schemas
Information
Schema:
strava_fitness_analysis
Object Info
Session
Strava Fitness Analysis
daily_activity
heartrate_daily
hourly_intensity
sleep_day
weight_log
Limit to 1000 rows
-- Categorize each BMI record
SELECT
BMI,
CASE
WHEN BMI < 18.5 THEN 'Underweight'
WHEN BMI >= 18.5 AND BMI < 25 THEN 'Normal'
WHEN BMI >= 25 AND BMI < 30 THEN 'Overweight'
ELSE 'Obese'
END AS BMI_Category
FROM weight_Log;
-- Count number of users in each BMI category
SELECT
BMI_Category,
COUNT(*) AS Count
FROM (
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
BMI BMI_Category
22.6499996185303 Normal
22.6499996185303 Normal
47.5400009155273 Obese
21.4500007629395 Normal
21.6900005340576 Normal
Result 6
Output
Action Output
Time Action Message
5 13:18:08 SELECT a.TotalSteps, s.TotalMinutesAsleep FROM daily_activity a JOIN sleep_day s ON a.Id = s.Id AND ... 410 row(s) returned
6 13:18:29 SELECT BMI, CASE WHEN BMI < 18.5 THEN 'Underweight' WHEN BMI >= 18.5 AND B... 67 row(s) returned

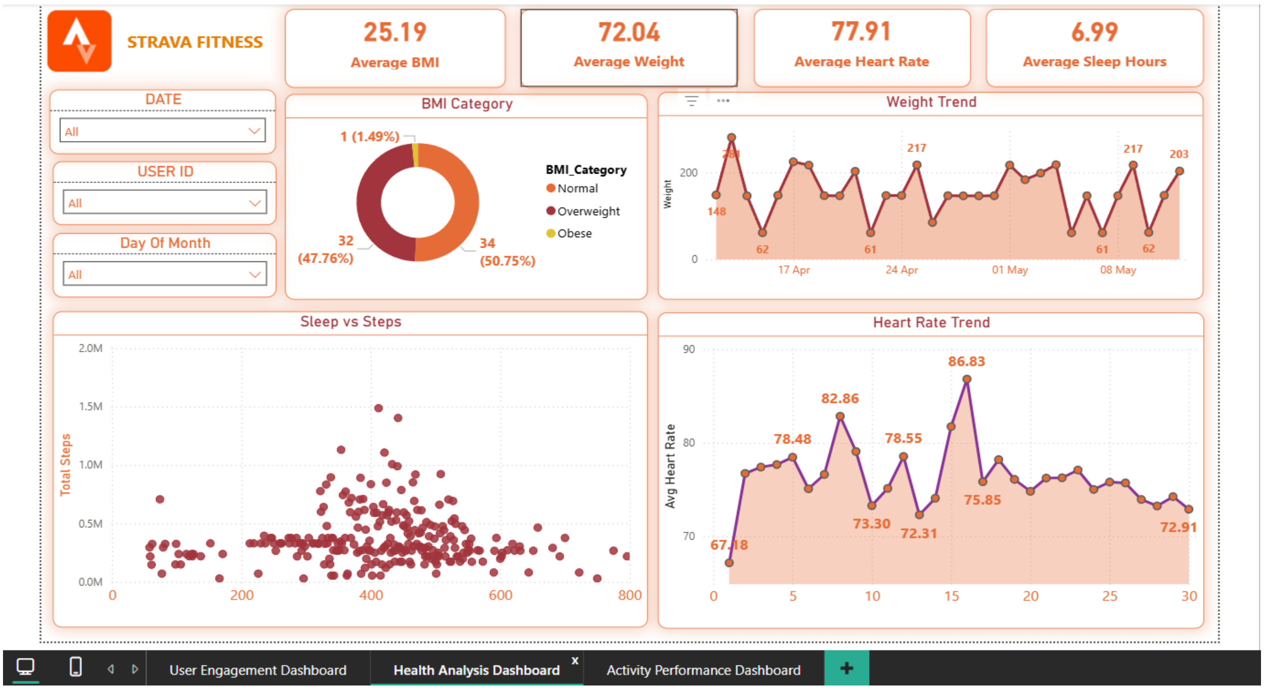


Power BI Dashboards –

Dashboard 1 — User Engagement



Dashboard 2 — Health Analysis



Dashboard 3 — Activity Performance